# Yue Dai

Assistant Professor · Department of Computer Science

*Illinois Institute of Technology, Chicago, IL 60616*

ydai21@illinoistech.edu

## Education

**University of Pittsburgh** *Pittsburgh, PA 15260*

PhD in Computer Science *2018/09 - 2025/06*

- Advisor: Dr. Youtao Zhang
- Co-Advisor: Dr. Xulong Tang

**University of Maryland, College Park** *College Park, MD 20742*

MS of Science in Telecommunication *2015/09 - 2017/05*

**Beihang University** *Beijing, China 100191*

Bachelor of Engineering in Electrical Engineering and Automation *2010/09-2014/06*

## Professional Experience

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|---|---|
| 2025/08-present | **Tenure-track Assistant Professor**, Department of Computer Science, Illinois Institute of Technology |
| 2022-2025 | **Graduate Research Assistant**, Department of Computer Science, University of Pittsburgh |
| 2018-2022 | **Graduate Teaching Assistant**, Department of Computer Science, University of Pittsburgh |
| 2016 | **Graduate Research Assistant**, Department of Computer Science, University of Maryland |
| 2014 | **Intern**, Information Department of Research Center of Automatic Control and Logistic Technology Engineering, Beijing Research Institute of Automation for Machinery Industry |
| 2014 | **Undergraduate Research Assistant**, Department of Electrical Engineering and Automation, Beihang University |
| 2013 | **Intern**, Department of Automatic System of Simons (China) LTD., Beijing Branch |

## Publications

**the authors contribute equally*

**Yue Dai***, Liang Liu*, Xulong Tang, Youtao Zhang, Jun Yang. 2025. MemFreezing: A Novel Adversarial Attack on Temporal Graph Neural Networks under Limited Future Knowledge. (The Forty-Second International Conference on Machine Learning (**ICML'2025**))

**Yue Dai**, Xulong Tang, Youtao Zhang. 2025. Cascade: A Dependency-Aware Efficient Training Framework for Temporal Graph Neural Network. 2025 ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'2025**)

Li, Yingheng, **Yue Dai**, Aditya Pawar, Rongchao Dong, Jun Yang, Youtao Zhang, and Xulong Tang. 2025. Using Reinforcement Learning to Guide Graph State Generation for Photonic Quantum Computers. The 52nd International Symposium on Computer Architecture (**ISCA'2025**).

Sheng Li, Qitao Tan, **Yue Dai**, Zhenglun Kong, Tianyu Wang, Jun Liu, Ao Li, Ninghao Liu, Yufei Ding, Xulong Tang, Geng Yuan. 2025. Mutual Effort for Efficiency: A Similarity-based Token Pruning for Vision Transformers in Self-Supervised Learning. The Thirteenth International Conference on Learning Representations (**ICLR'2025**)

**Yue Dai**, Youtao Zhang, Xulong Tang. 2023. CEGMA: Coordinated elastic graph matching acceleration for graph matching networks. 2023 IEEE International Symposium on High-Performance Computer Architecture (**HPCA'2023**).

**Yue Dai**, Xulong Tang, Youtao Zhang. 2023. FlexGM: An Adaptive Runtime System to Accelerate Graph Matching Networks on GPUs. 2023 IEEE 41st International Conference on Computer Design (**ICCD'2023**).

Sheng Li*, Geng Yuan*, **Yue Dai***, Youtao Zhang, Yanzhi Wang, Xulong Tang. 2023. Smartfrz: An efficient training framework using attention-based layer freezing. The 11th International Conference on Learning Representations (**ICLR'2023 *Spotlight***).

**Yue Dai**, Xulong Tang, Youtao Zhang. 2022. An efficient segmented quantization for graph neural networks. CCF Transactions on High Performance Computing, 4(4), 461-473.(**THPC'2022**)

Zhexiong Liu*, Meiqi Guo*, **Yue Dai***, Diane Litman. 2022. ImageArg: A multi-modal tweet dataset for image persuasiveness mining. Proceedings of the 9th Workshop on Argument Mining, International Conference on Computational Linguistics at **COLING'2022**.

Yuting Wu, Mingxing Wang, Xiu Jin, **Yue Dai**, Liang Liu, Wenjie Yuan, Yong Ye, Zhengkang Zhu, Lichuan Gu, Youtao Zhang. 2026. AgroPatch AutoStrike for Adversarial Patch Attacks on Plant Disease Detection Models. **iScience**, 29(8).

Haroon Gharwi, Kai Shu, **Yue Dai**. 2026. AnchorRM: Parallel Intra-Batch State Reconstruction with Anchor-Based Residual Memory for Temporal Graph Neural Networks. arXiv preprint arXiv:2610.27558.

**Yue Dai**, Sarah Zhou, Youtao Zhang. 2026. RAFT: Redundancy-Aware Input Filtering for Temporal Graph Neural Network Training. arXiv preprint arXiv:2610.24875.

Yang Zhao*, **Yue Dai***, Liang Qiao, Buxin Tu, Bo Liu, Sen Zhang, Yi Zhang, Xiaoyu Hao, Jun Shi, Junshi Chen, Hong An. 2026. A Dependency-Aware Multi-GPU Training Framework for Temporal Graph Neural Networks. arXiv preprint arXiv:2610.15730.

Haroon Gharwi, **Yue Dai**, Kai Shu. 2026. StateFlow: Dual-State Recurrent Modeling for Long-Horizon Time Series Forecasting. arXiv preprint arXiv:2607.00197.

Sheng Li, Yang Sui, Junhao Ran, Bo Yuan, **Yue Dai**, and Xulong Tang. Temporal Aware Pruning for Efficient Diffusion-based Video Generation. arXiv preprint arXiv:2605.17837 (2026).

Sheng Li, Yang Sui, Yue Wu, Zhuoran Song, Bo Yuan, Xulong Tang, and **Yue Dai**. "Accelerating 3D Gaussian Splatting using Tensor Cores." arXiv preprint arXiv:2605.17855 (2026).

Sheng Li, Geng Yuan, Yawen Wu, **Yue Dai**, Chao Wu, Alex K Jones, Jingtong Hu, Yanzhi Wang, Xulong Tang. 2024. EdgeOL: Efficient in-situ Online Learning on Edge Devices. arXiv preprint arXiv:2401.16694.

Tianyu Wang, Sheng Li, Bingyao Li, **Yue Dai**, Ao Li, Geng Yuan, Yufei Ding, Youtao Zhang, Xulong Tang. 2024. Improving GPU multi-tenancy through dynamic multi-instance GPU reconfiguration. arXiv preprint arXiv:2407.13126.

Justin Brody, Samuel Barham, **Yue Dai**, Christopher Maxey, Donald Perlis, David Sekora, Jared Shamwell. 2016. Reasoning with grounded self-symbols for human-robot interaction. 2016 AAAI Fall Symposium Series

## Research Experience

**Illinois Institute of Technology** *Chicago, IL*

Assistant Professor *2025-Present*

- Develop efficient training frameworks for temporal graph neural networks through input filtering and system optimizations
- Investigate efficient GraphRAG through offline graph compression and adaptive online graph expansion
- Investigate efficient graph-based memory for multi-agent systems through memory refinement and filtering
- Investigate efficient agentic image and video generation through workflow optimization, feature caching and rerouting
- Study adversarial robustness, error propagation, and efficient recovery in graph-based multi-agent systems

**University of Pittsburgh** *Pittsburgh, PA*

Research Assistant *2018-2025*

- Develop efficient training frameworks for general deep neural networks and graph neural networks
- Design software-hardware co-designs and GPU runtimes for inference acceleration on diverse deep graph learning models
- Develop adversarial attacks and defenses for deep graph learning models, with a focus on temporal graph neural networks
- Optimize deep graph learning models for accurate and scalable graph similarity computing
- Develop graph-based reinforcement learning methods for efficient photonic quantum computing

**University of Maryland, College Park** *College Park, MD*

Research Assistant *2015-2017*

- Develop metacognitive intelligent system based on active logic machine
- Design a reliable and secure distributed data management method for cloud services

**Beihang University** *Beijing, CHINA*

Research Assistant *2010-2014*

- Design sensor and control system for solar panel maintaining robots and devices

**Beijing Research Institute of Automation for Machinery Industry** *Beijing, CHINA*

Research Intern *2014*

- Develop an electrical nonlinearity pre-compensation method for optical orthogonal frequency-division multiplexing systems

## Research Talks

2025 **Efficient and Robust Dynamic Deep Graph Learning**, at University of California at Riverside, Riverside, USA

2025 **Towards Efficient and Robust Deep Graph Learning**, at University of Pittsburgh, Pittsburgh, USA

2023 **An Adaptive Runtime System to Accelerate Graph Matching Networks on GPUs**, at ICCD 2023, Washington. DC, USA

2020 **Effectiveness of Video Encoder for Adversarial Videos Defense**, at University of Pittsburgh, Pittsburgh, PA.

## Awards

2024 **CS 50 Outstanding Research Fellowship**, Department of Computer Science, University of Pittsburgh

2023 **Orrin E. and Margaret M. Taulbee Graduate Award**, Department of Computer Science, University of Pittsburgh

## Mentoring

2026-Now **Chenxi Gao**, PhD Student, Illinois Institute of Technology

2026-Now **Haroon Gharwi**, PhD Student, Illinois Institute of Technology

2025-2026 **Sebastian Alejandro Mendoza**, Master of Science, Illinois Institute of Technology

2021-2025 **Sheng Li**, PhD Student, University of Pittsburgh

2022-2025 **Yingheng Li**, PhD Student, University of Pittsburgh

2021-2022 **Austin Tercha**, Master of Science, University of Pittsburgh

## Teaching

Fall 26 **CS577 DEEP LEARNING**, Instructor

Spring 26 **CS595 EFFICIENT MACHINE LEARNING SYSTEM**, Instructor

Fall 22 **CS2210 COMPILER DESIGN**, Teaching Assistant

Fall 22 **CS1622 INTRO TO COMPILER DESIGN**, Teaching Assistant

Summer 22 **CS0007 INTRO TO COMPUTER PROGRAMMING**, Teaching Assistant

Spring 22 **CS1550 INTRO TO OPERATING SYSTEMS**, Teaching Assistant

Summer 21 **CS1501 ALGORITHMS DATA STRUCTURES 2**, Teaching Assistant

FALL 20 **CS2510 COMPUTER OPERATING SYSTEMS**, Teaching Assistant

FALL 20 **CS1621 STRUCTURE PROGRAMMING LANGUAGES**, Teaching Assistant

Summer 20 **CS0007 INTRO TO COMPUTER PROGRAMMING**, Teaching Assistant

Spring 20 **CS1550 INTRO TO OPERATING SYSTEMS**, Teaching Assistant

Fall 19 **CS0008 INTRO TO COMPUTER PROGRAMMING WITH PYTHON**, Teaching Assistant

Summer 19 **CS0007 INTRO TO COMPUTER PROGRAMMING**, Teaching Assistant

Spring 19 **CS1520 PROGRAMMING LANGUAGE FOR WEB APPLICATIONS**, Teaching Assistant

Fall 18 **CS0449 INTRO TO SYSTEMS SOFTWARE**, Teaching Assistant

## Professional Service

**Registration Chair** ASPLOS'2026

**Program Committee** ISCA'26, MLSys'26, MICRO'26, HPCA'26, EuroSys'27, HPCA'27

**Artifact Evaluation Committee** MICRO'22, ASPLOS'23, PLDI'23,

**Peer Review** ICLR'24, ICLR'25, ICLR'26, MLSys'26, ISCA'26, TCC'26, J. Supercomput'26, Clust. Comput'26, TACO'26, TC'26, ICLR'27

**Guest Editor** IEEE Internet Computing'26

**Others** NSF Reviewer 2026